Cisco Packet Tracer

Level 4 – DHCP (Dynamic Host Configuration Protocol)

Cisco Packet Tracer | Topology: 1 Router + 2 Switches + 12 PCs

Overview

This level builds on everything from Levels 2 and 3 – VLANs, trunking, and inter-VLAN routing are all configured the same way. The key difference is that IP addresses are no longer assigned manually to each PC. Instead, DHCP (Dynamic Host Configuration Protocol) handles that automatically.

In a real company with hundreds of devices, manually assigning IP addresses to every single machine is impractical and error prone. DHCP eliminates that by automatically assigning the correct IP address, subnet mask, default gateway, and DNS server to any device that connects to the network.

Why Switches and Routers Work Together

Understanding the role of each device is important before getting into DHCP configuration.

A switch acts like a toll gate – it allows and rejects traffic based on their VLAN membership. Every packet that comes into a switch is checked against its VLAN assignment. Without this, every packet from every device floods the entire network simultaneously, causing congestion, security vulnerabilities, and in extreme cases a broadcast storm where so much traffic is flying everywhere that the network chokes itself completely.

A router acts as the decision maker. It knows all the subnets, decides where traffic goes, and in this level, also manages IP assignment through DHCP. Without the router, VLANs remain completely isolated with no way to communicate and no way to get IP addresses automatically.

Together they create a structured network – the switch enforces the boundaries between departments, while the router manages controlled communication across those boundaries and handles automatic IP assignment for every device that joins the network.

What is DHCP?

DHCP stands for Dynamic Host Configuration Protocol. It is a layer 3 function – meaning it works at the network layer where IP addresses and routing decisions are made. When a PC connects to the network and is set to DHCP mode, it sends out a broadcast message essentially saying, ‘I just joined the network, can someone give me an IP address?’

The router receives this request, checks which VLAN the PC is on, looks up the corresponding DHCP pool, and automatically assigns:

- **IP Address –** a unique address for that device on the network
- **Subnet Mask –** defines which network the devices belongs to
- **Default Gateway –** the router subinterface IP, so the devices know where to send traffic outside its VLAN
- **DNS Server –** for resolving domain names to IP addresses

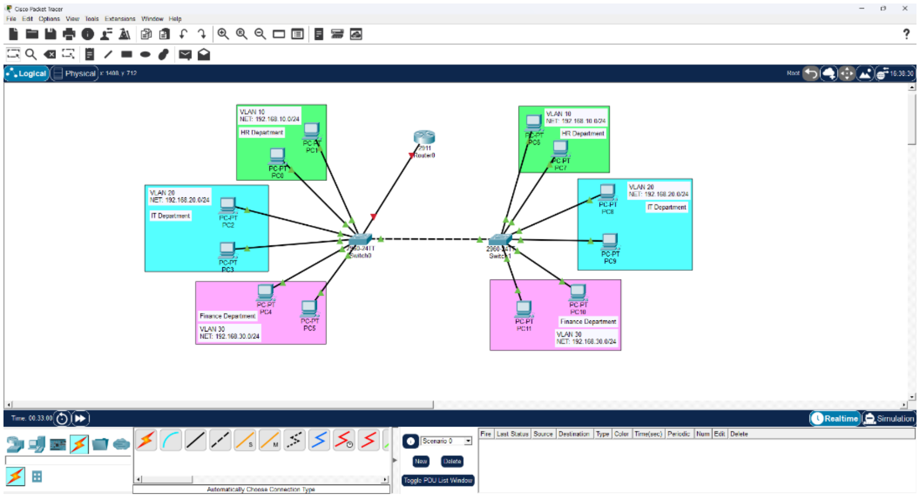
NOTE: *DHCP is a Layer 3 function because it deals with IP addressing, which is a network layer concern. Even though the initial request starts as a Layer 2 broadcast, the router (Layer 3 device) is the one that processes it and makes the assignment decision.*

Configuration Path

- Step 1 - Design and label the network topology
- Step 2 – Configure VLANs, assign ports, and set up trunk links
- Step 3 – Configure router subinterfaces (Router-on-a-Stick)
- Step 4 – Configure DHCP pools on the router
- Step 5 – Exclude reserved IP address ranges
- Step 6 – Set PCs to DHCP mode and verify

Step 1 – Design the Network Topology

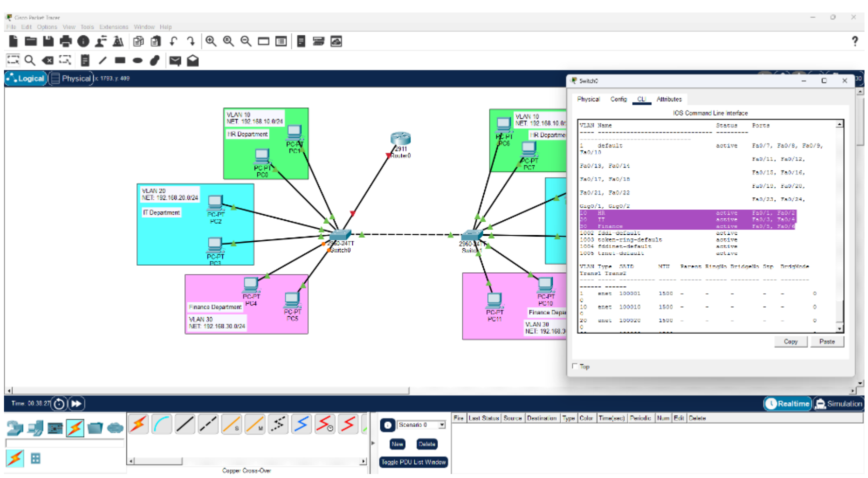
Build the same topology used in Level 3 – two switches, one router, and 12 PCs across three departments. Label everything clearly: VLAN numbers, department names, subnet ranges, and router subinterface labels.



Good labeling at this stage saves a lot of confusion later. You should be able to look at the topology and immediately know which VLAN each device belongs to and what IP range that VLAN uses.

Step 2 – Configure VLANs, Ports, and Trunk links

This step is identical to Level 2 – create and name the VLANs, assign the switch ports to their correct VLANs, and configure trunk links between the switch and router.



The only difference from previous levels – do not assign IP addresses to the PCs yet. Leave them on Static with empty fields for now. DHCP will handle that in Step 6.

On Both Switches – Create VLANs

```
en
config t
vlan 10
name HR
exit
vlan 20
name IT
exit
vlan 30
name Finance
exit
```

Assign Ports to VLANs

```
int range fa0/1-2
switchport mode access
switchport access vlan 10
exit
int range fa0/3-4
switchport mode access
switchport access vlan 20
exit
```

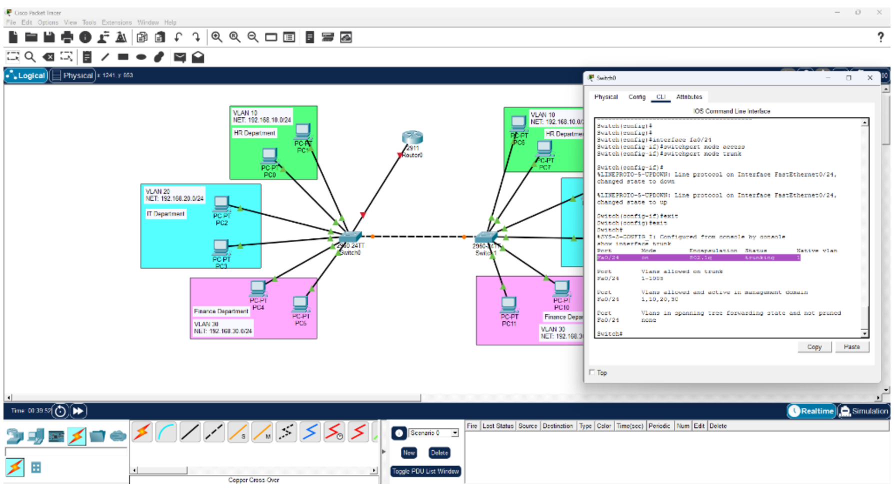
int range fa0/5-6

switchport mode access

switchport access vlan 30

exit

Configure Trunk Links



Switch to switch trunk (Fa0/24 on both switches):

int fa0/24

switchport mode trunk

exit

Switch to router trunk (Gig0/1 on Switch0)

int gig0/1

switchport mode trunk

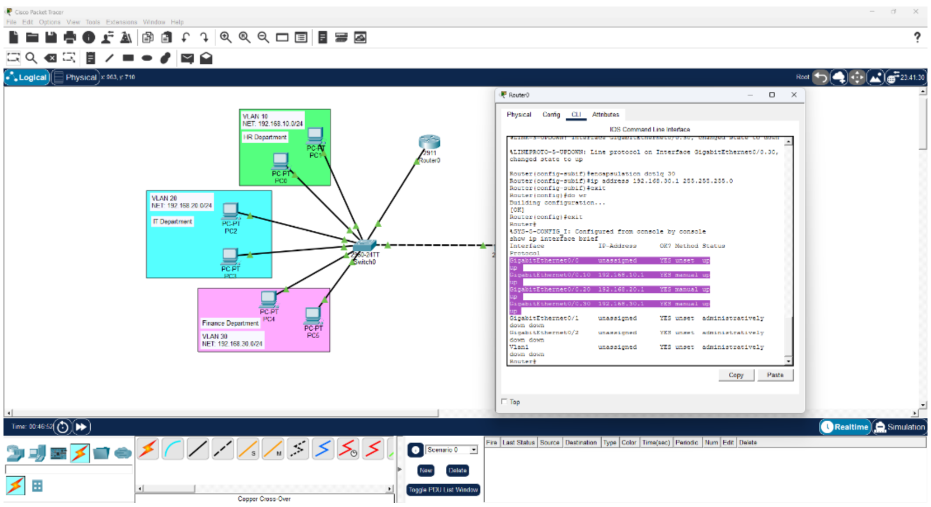
exit

do wr

IMPORTANT: *Don't forget to trunk BOTH the switch-to-switch AND switch-to-router ports. This was the most common mistakes in Level 3 – the router port being left in access mode. Always verify with 'show interface trunk' after configuring.*

Step 3 – Configure Router Subinterfaces

Same as level 3 – configure subinterfaces on the router for each VLAN using encapsulation dot1q. These subinterfaces IPs will also serve as the default gateways assigned by DHCP.




```
en

config t

int gig0/0

no shutdown exit

int gig0/0.10

ecapsualtion dot1q. 10

ip address 192.168.10.1 255.255.255.0

exit

int gig0/0.20

encapsulation dot1q 20
```

ip address 192.168.20.1 255.255.255.0

exit

int gig0.0/30

encapsulation dot1q 30

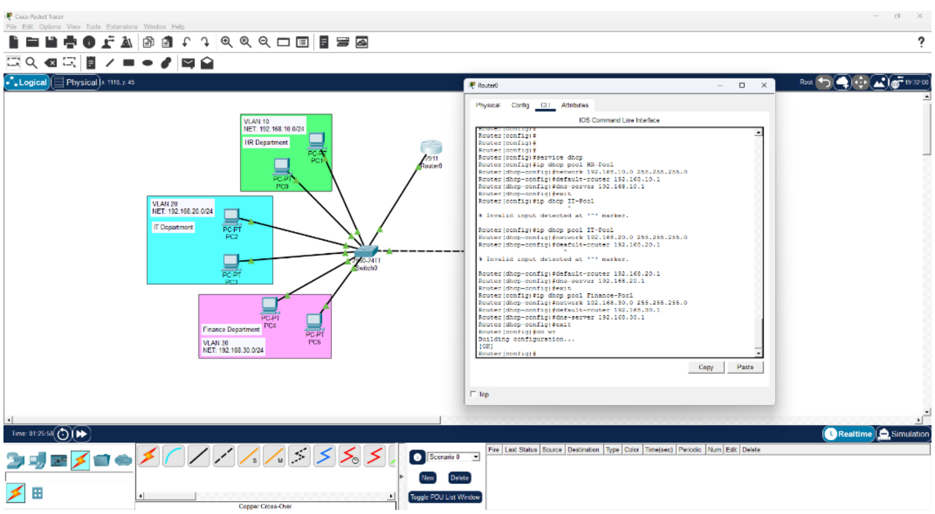
ip address 192.168.30.1 255.255.255.0

exit

do wr

Step 4- Configure DHCP Pools

This is the main new configuration in this level. DHCP pools are created on the router – one per VLAN. Each pool defines the network range the router will assign IPs from for that specific VLAN.



What is DHCP Pool?

A DHCP pool is defined range of IP addresses that the router is allowed to hand out to devices on a specific network. Think of it like a pool of available addresses – when a device connects and requests an IP, the router dips into that pool and assigns one. When that device disconnects, the address goes back into the pool to be reused.

Each VLAN needs its own separate pool because each VLAN is on a different subnet. DHCP needs to know which range of addresses belongs to which network so its assigns the right IP to the right device.

Enable DHCP Service
First enable the DHCP service on the router:

```
service dhcp
```

This activates the DHCP server function on the router. Without this command, the router will not respond to DHCP requests from PCs even if pools are configured.

Create DHCP Pools for Each VLAN

HR Department – VLAN 10:

```
ip dhcp pool HR-Pool
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 192.168.10.1
exit
```

IT Department – VLAN 20:

```
ip dhcp pool IT-Pool
network 192.168.20.0 255.255.255.0
default-router 192.168.20.1
dns-server 192.168.20.1
exit
```

Finance Department – VLAN 30:

```
ip dhcp pool Finance-Pool
network 192.168.30.0 255.255.255.0
```

```
default-router 192.168.30.1
dns-server 192.168.30.1
exit
do wr
```

Understand the Commands

- **Ip dhcp pool [name] –** creates a new DHCP pool with the given name and enters pool configuration mode
- **Network [address] [mask] –** defines which subnet this pool servers. The router will only assign IPs within this range to devices on this network
- **Default-router [IP] –** tells the PC what its default gateway is. This is automatically sent to the PC along with the IP address – no need to manually set it anymore
- **Dns-server [IP] –** tells the PC which DNS server to use for resolving domain names

NOTE: *The dns-server is set to the same IP as the default gateway (the router subinterface) because in this lab, the router is handling both routing and DNS resolution. In a real enterprise environment, the DNS server would typically be a dedicated server with its own IP address – like a Windows Server running DNS, or a public DNS like 8.8.8.8 (Google) or 1.1.1.1 (Cloudflare). The reason we don't use those here is that our lab network is isolated with no actual internet connection. Those public DNS servers wouldn't be reachable anyway.*

Verify DHCP Pools

To confirm the pools are configured correctly, run:

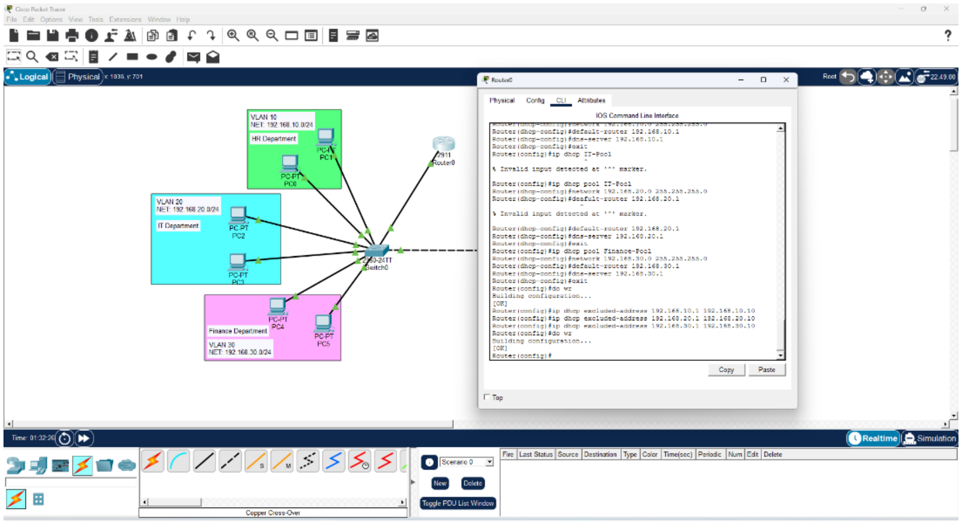
```
show ip dhcp pool
```

To check which Ip addresses have been assigned:

```
show ip dhcp binding
```

Step 5 – Exclude Reserved IP Addresses

Before DHCP starts handing out addresses, certain IPs need to be excluded from the pool. These are addresses that are already assigned to specific devices – like the router subinterface IPs – or addresses reserved for future use like servers, printers, or network equipment.



If exclusions are not set, DHCP might assign the gateway IP (192.168.10.1) to regular PC -which could cause conflict and break connectivity for everyone on that VLAN. This is why exclusions are configured before any PC gets an IP.

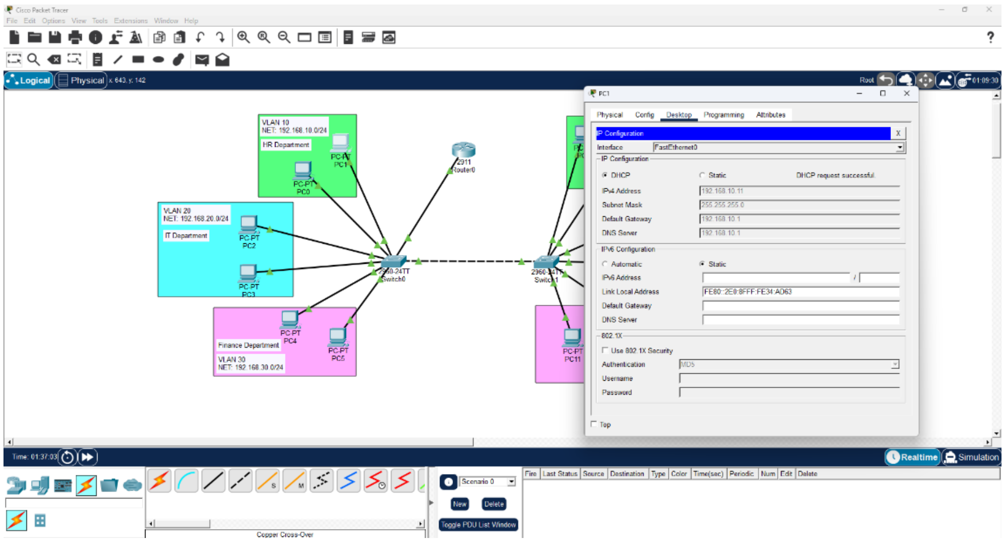
The exclusion range defines a block of addresses DHCP will never assign dynamically. In this lab the first 10 addresses of each subnet are excluded – reserving them for network infrastructure and any future static assignments:

```
ip dhcp excluded-address 192.168.10.1 192.168.10.10
ip dhcp excluded-address 192.168.20.1 192.168.20.20
ip dhcp excluded-address 192.168.30.1 192.168.30.30
do wr
```

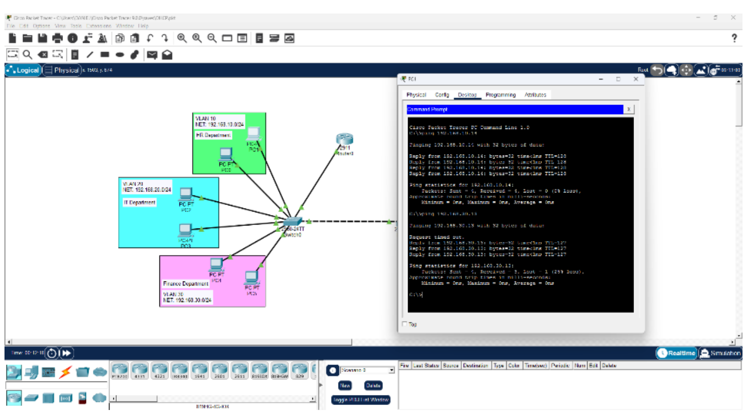
BEST PRACTICE: *Excluding a range of IPs at the start of each subnet (.1 to .10) is a common real-world practice. The .1 address is always the gateway, .2 through .10 are kept available for static assignment to infrastructure devices like servers, printers, or access points. DHCP then assigns address starting from .11 onwards to regular client devices.*

Step 6 – Set PCs to DHCP and Verify

With all the router configurations done, the last step is to switch each PC from Static to DHCP mode and verify they receive the correct IP address automatically.



1. Click on any PC and go to **Desktop > IP Configuration**.
2. Select **DHCP** instead of Static.
3. After a moment the PC should automatically populate its IP address, subnet mask, default gateway and DNS server – all from the DHCP pool on the router.
4. Verify the assigned IP is in the correct range for that VLAN – HR PC's should get 192.168.10.1 or higher, IT PCs 192.168.20.11 or higher, Finance PCs 192.168.30.11 or higher.
5. Repeat for all 12 PCs across both switches.
6. Once all PCs have IPs, test cross-VLAN connectivity with ping to confirm everything is working.



NOTE: *The first IP assigned by DHCP will be the first available address after exclusion range. Since .1 through .10 are excluded, the first PC to connect will receive .11, the next .12, and so on.*

Troubleshooting – DHCP Failed

After switching a PC to DHCP mode, instead of receiving an IP address automatically, it showed a 'DHCP Failed' error. Here's what happened and how it was fixed.

What Happened

The DHCP request from the PC was note reaching the router. The PC sent out a broadcast asking for an IP address, but the router never responded.

How it was Indentified

After checking the DHCP pool configuration on the router – everything looked correct. The issue had to be somewhere between the PC and the router. Running show interfaces trunk on the switch revealed the router port was missing from the trunk list – same issue as Level 3.

Root Cause

The switch port connected to the router was not configured as a trunk. Since this was a fresh topology rebuild, the trunk configuration on the router port was missed during setup.

DHCP requests are broadcasts – they start at Layer 2 and travel through the switch. If the port between the switch and router is not trunk, the VLAN-tagged DHCP broadcast never reaches the router. The router sits there with perfectly configured DHCP pools but never receives any requests to respond to.

Fix

```
int gig0/1
switchport mode trunk
exit
do wr
```

After trunking the router port, switching any PC to DHCP mode immediately assigned the correct IP address from the right pool.

DISCOVERY: *This is now the second time the missing router trunk caused a failure – first in Level 3 with inter-VLAN routing, and again here with DHCP. The lesson is reinforced; any port carrying multiple VLANs must be trunk – including switch port connected to the router. Always verify with 'show interface trunk' before testing.*

Key Takeaways

- **DHCP is a Layer 3 function –** it deals with IP addressing and is configured on the router, not the switch
- **A DHCP pool defines the address range per VLAN –** each VLAN needs its own pool matching its subnet
- **Always enable DHCP first –** 'service dhcp' must be run before any pools will respond to requests
- **Exclude reserved addresses before testing –** gateway IPs and infrastructure addresses must be excluded or DHCP may assign them to regular PCs causing conflicts
- **The router trunk must still be configured –** DHCP broadcasts travel through the switch to the router. If the router port is not a trunk, DHCP requests never arrive
- **Show Ip dhcp pool –** verifies pool configuration
- **Show IP dhcp binding –** shows which IPs have been assigned and to which MAC address
- **Dns-server uses the gateway IP in this lab –** in a real environment, it would be a dedicated DNS server or public DNS. The lab network is isolated so public DNS servers like 8.8.8.8 are not reachable.